

Grade 9 Science Curriculum

U.S. Standards-Aligned Course Syllabi

BIOLOGY

Life Science
36 weeks • 4 modules

CHEMISTRY

Physical Science
36 weeks • 4 modules

PHYSICS

Physical Science
36 weeks • 4 modules

Curriculum Architecture

Each course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

The U.S. does not have one national Grade 9 science sequence. This curriculum uses the NGSS high-school 9–12 framework as the primary national reference and cross-checks major themes against state frameworks including California NGSS and Texas high-school science TEKS. Course placement varies by district, so these are presented as U.S. Standards-Aligned Grade 9 courses.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Questions should emphasize explanation, modeling, data interpretation, quantitative reasoning, application to unfamiliar phenomena, and evidence-based reasoning rather than memorization alone. Each module assessment should sample all major concepts and include scenario/data/model items.

This 36-week course is designed around **HS-LS1, HS-LS2, HS-LS3, HS-LS4**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	9
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Explain systems from molecules and cells through populations and ecosystems.
- Use models, data and evidence to explain biological phenomena.
- Analyze inheritance, gene expression, evolution and ecological interactions.
- Conduct and evaluate scientific investigations.
- Communicate scientific explanations with accurate terminology and reasoning.

Biology

Module 1 — Foundations of Biology: Chemistry, Cells & Organization

Week	Topics	Students will be able to	Standards
Week 1 — Biology & Scientific Investigation	Characteristics of life; observations vs. inferences; hypotheses; variables; controls; experimental design; models	Identify variables and controls; construct evidence-based conclusions	SEP: questions, investigations, data; CCC: cause/effect
Week 2 — Chemistry of Life	Matter; elements; atoms; isotopes; ions; ionic/covalent bonds; polarity	Describe atomic structure; relate valence electrons to behavior	HS-PS1 foundation
Week 3 — Water & Biological Molecules	Water polarity; hydrogen bonding; pH; buffers; carbohydrates; lipids; proteins; nucleic acids	Classify macromolecules; connect structure to function	Structure/function; matter and energy
Week 4 — Cell Structure & Function	Cell theory; prokaryotic/eukaryotic; plant/animal cells; organelles; specialization	Compare cell types; explain organelle functions	HS-LS1 structure/function
Week 5 — Membranes & Transport	Selective permeability; diffusion; osmosis; facilitated diffusion; active transport; endo/exocytosis	Predict membrane movement; explain homeostasis	HS-LS1 interacting systems
Week 6 — Energy & Enzymes	ATP; activation energy; enzymes; active sites; temperature; pH; concentration	Analyze enzyme data; connect ATP to cellular work	HS-LS1 energy/matter
Week 7 — Cellular Respiration	Glycolysis; Krebs cycle; electron transport; aerobic/anaerobic respiration; fermentation	Trace energy transfer; compare pathways	HS-LS1 energy/matter
Week 8 — Photosynthesis	Chloroplasts; pigments; light reactions; Calvin cycle; carbon fixation	Model photosynthesis; connect to respiration	HS-LS1/LS2 energy
Week 9 — Module Assessment	Integrated review of chemistry, cells, transport, enzymes, respiration and photosynthesis	Synthesize concepts; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

Biology

Module 2 — Genetics, DNA & Cellular Reproduction

Week	Topics	Students will be able to	Standards
Week 1 — Cell Cycle & Mitosis	Cell cycle; interphase; chromosomes; mitosis; cytokinesis; cancer	Model division; explain growth; connect uncontrolled division to cancer	HS-LS1 cell division
Week 2 — Meiosis & Sexual Reproduction	Meiosis; haploid/diploid; gametes; homologous chromosomes; crossing over; assortment	Compare mitosis/meiosis; explain variation	HS-LS3 inheritance
Week 3 — DNA Structure & Replication	Nucleotides; double helix; base pairing; genes; chromosomes; replication	Model DNA; explain replication	HS-LS3 information
Week 4 — RNA & Transcription	mRNA; tRNA; rRNA; transcription; codons; RNA processing	Explain transcription; interpret codons	HS-LS1/LS3 information flow
Week 5 — Protein Synthesis	Translation; ribosomes; amino acids; DNA → RNA → protein	Model translation; connect DNA to proteins	HS-LS1-1
Week 6 — Mendelian Genetics	Alleles; genotype/phenotype; dominant/recessive; Punnett squares; probability	Solve monohybrid crosses; predict inheritance	HS-LS3
Week 7 — Beyond Mendel	Incomplete dominance; codominance; multiple alleles; sex-linked; polygenic traits	Analyze non-Mendelian patterns	State biology framework extension
Week 8 — Mutations & Biotechnology	Substitution; insertion/deletion; frameshift; disorders; genetic engineering; ethics	Classify mutations; predict effects; evaluate biotech scenarios	HS-LS3 variation
Week 9 — Module Assessment	Cell division, DNA/RNA, proteins, inheritance, variation	Integrate genetics; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

Biology

Module 3 — Evolution, Natural Selection & Biological Diversity

Week	Topics	Students will be able to	Standards
Week 1 — Evidence for Evolution	Fossils; anatomy; molecular evidence; embryology; biogeography	Evaluate lines of evidence; infer ancestry	HS-LS4 evidence
Week 2 — Natural Selection	Variation; heritability; competition; selection pressure; differential reproduction	Explain population change and adaptation	HS-LS4 natural selection
Week 3 — Evolutionary Mechanisms	Mutation; gene flow; drift; founder/bottleneck; recombination; allele frequencies	Compare mechanisms; interpret population data	HS-LS4 mechanisms
Week 4 — Speciation	Species; geographic/behavioral/temporal isolation; divergence; adaptive radiation	Model speciation; identify barriers	HS-LS4 species formation
Week 5 — Classification & Phylogeny	Taxonomy; domains; cladograms; phylogenetic trees; derived traits	Read phylogenies; infer relatedness	Modeling/evidence
Week 6 — Biodiversity	Genetic/species/ecosystem diversity; extinction; conservation	Analyze threats; connect diversity and stability	HS-LS2/LS4
Week 7 — Adaptation & Coevolution	Structural/behavioral/physiological adaptations; mimicry; predator-prey; host-parasite	Analyze adaptations and coevolution	HS-LS4
Week 8 — Evolution in Action	Antibiotic resistance; pesticide resistance; viral evolution; artificial selection	Analyze real-world evolution data	HS-LS4 applications
Week 9 — Module Assessment	Evolutionary evidence, selection, mechanisms, speciation, biodiversity	Synthesize; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

Biology

Module 4 — Ecology, Ecosystems & Sustainability

Week	Topics	Students will be able to	Standards
Week 1 — Ecological Organization	Organism; population; community; ecosystem; biome; biosphere; habitat; niche	Identify levels; distinguish habitat/niche	HS-LS2
Week 2 — Energy Flow	Producers; consumers; decomposers; food webs; trophic levels; energy pyramids	Trace energy; interpret food webs	HS-LS2 energy transfer
Week 3 — Matter Cycling	Water; carbon; nitrogen; phosphorus; decomposition; human impacts	Model cycles; explain conservation of matter	HS-LS2 matter cycling
Week 4 — Population Ecology	Density; distribution; birth/death; migration; exponential/logistic growth; carrying capacity	Model population growth; interpret data	HS-LS2 population dynamics
Week 5 — Community Ecology	Competition; predation; herbivory; parasitism; mutualism; commensalism; niche	Classify interactions; predict outcomes	HS-LS2 interactions
Week 6 — Ecosystem Dynamics	Disturbance; succession; resilience; limiting factors; feedback	Model ecosystem change; evaluate resilience	HS-LS2 stability/change
Week 7 — Human Impact	Habitat loss; pollution; climate change; invasive species; overharvesting	Analyze environmental impacts and evidence	HS-LS2/ESS3
Week 8 — Sustainability & Conservation	Renewable resources; agriculture; water; energy; restoration; conservation	Evaluate solutions; propose evidence-based actions	Engineering/Earth-s system connections
Week 9 — Final Assessment	Full-course synthesis: molecules → cells → organisms → populations → ecosystems	Solve unfamiliar scenarios; demonstrate mastery	Final performance assessment

Module delivery notes. Use diagrams and models heavily. Favor cause-and-effect reasoning, biological data interpretation and real-world phenomena such as disease, resistance, biodiversity and ecosystem change.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	HS-LS1, HS-LS2, HS-LS3, HS-LS4
Science & Engineering Practices	Questioning; modeling; investigation; mathematics / computational thinking; data analysis; explanation; evidence; design
Crosscutting Concepts	Patterns; cause/effect; systems; energy/matter; structure/function; stability/change
State mapping	California NGSS and Texas high-school science frameworks can be mapped at chapter / question level without changing the core 36-week structure.

Content Production Guidance

For each week, expand the syllabus topic into the established LucidPrep package: ~2-minute opening, objectives, vocabulary, 5 warm-ups with explanations, lecture, 2 ~2-minute videos, AI Instructor presentation, two homework sets of 5, 40-question assessment, answer explanations and mastery/retake handling. Build question banks around phenomena, scenarios, data, models and application.

This 36-week course is designed around **HS-PS1, HS-PS3 + engineering connections**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	9
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Explain matter and properties using atomic and molecular models.
- Use periodic patterns to reason about element behavior and bonding.
- Represent, balance and quantitatively analyze reactions.
- Apply moles, stoichiometry, gases, concentration and energy.
- Evaluate chemistry applications and environmental phenomena.

Chemistry

Module 1 — Matter, Measurement & Atomic Foundations

Week	Topics	Students will be able to	Standards
Week 1 — Chemistry & Scientific Investigation	Safety; SI units; measurement; accuracy/precision; significant figures; graphs	Design safe investigations; communicate quantitative evidence	SEP; Texas Chemistry practices
Week 2 — Matter & Properties	States; physical/chemical properties; changes; density	Classify properties/changes; calculate density	HS-PS1
Week 3 — Atomic Structure	Protons; neutrons; electrons; atomic/mass number; isotopes; ions	Model atoms; determine particle counts	HS-PS1
Week 4 — Electron Structure & Periodic Patterns	Energy levels; valence electrons; electron configurations; periodic trends	Predict properties from periodic position	HS-PS1-1
Week 5 — Chemical Bonding	Ionic; covalent; metallic; Lewis structures; polarity; electronegativity	Model bonding; connect bonding to properties	HS-PS1-1/2/3
Week 6 — Molecular Structure & Intermolecular Forces	Geometry; polarity; intermolecular forces; solubility; boiling/melting	Relate structure to properties	HS-PS1
Week 7 — Chemical Formulas & Naming	Ionic/molecular compounds; formula writing; naming; composition	Write/interpret formulas; calculate simple composition	HS-PS1 quantitative
Week 8 — Matter Models & Review	Particle models; macro/micro representations; evidence	Connect models to observations	Modeling/evidence
Week 9 — Module Assessment	Atoms; periodic trends; bonding; molecular structure	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

Chemistry

Module 2 — Chemical Reactions & Quantitative Chemistry

Week	Topics	Students will be able to	Standards
Week 1 — Chemical Reactions	Reactants/products; reaction evidence; conservation of matter	Represent reactions; explain conservation	HS-PS1
Week 2 — Balancing Equations	Coefficients; conservation of atoms	Balance equations and justify coefficients	HS-PS1-7
Week 3 — Reaction Types	Synthesis; decomposition; replacement; combustion; acid-base	Classify and predict simple reactions	Chemistry course foundations
Week 4 — Mole & Molar Mass	Mole; Avogadro; molar mass; conversions	Convert particles ↔ moles ↔ mass	HS-PS1 quantitative
Week 5 — Stoichiometry	Mole ratios; mass-mass; limiting reactant; yield	Solve stoichiometry; evaluate yield	Quantitative chemistry
Week 6 — Aqueous Reactions	Solutions; dissociation; solubility; precipitation; ionic reasoning	Interpret aqueous reactions	HS-PS1
Week 7 — Acids & Bases	Acids; bases; pH; indicators; neutralization; strong/weak	Interpret pH; predict neutralization	HS-PS1
Week 8 — Rates & Equilibrium Foundations	Collision model; temperature; concentration; surface area; catalysts; equilibrium	Interpret rate data; explain catalysts	HS-PS1/PS3
Week 9 — Module Assessment	Reactions; equations; moles; stoichiometry; solutions; acids/bases; rates	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

Chemistry

Module 3 — Gases, Energy & Chemical Systems

Week	Topics	Students will be able to	Standards
Week 1 — Gas Behavior	Pressure; volume; temperature; particle motion	Predict gas behavior	PS1/PS3 modeling
Week 2 — Gas Laws	Boyle; Charles; combined gas law; ideal gas concept	Solve basic gas-law problems	Quantitative modeling
Week 3 — Thermochemistry	Exo/endothermic; heat; temperature; calorimetry; energy diagrams	Interpret/calculate energy changes	HS-PS1/PS3
Week 4 — Chemical Kinetics	Collision theory; activation energy; rate factors; catalysts	Analyze rate data	HS-PS1
Week 5 — Chemical Equilibrium	Dynamic equilibrium; Le Châtelier; concentration/temperature effects	Predict qualitative shifts	HS-PS1 systems
Week 6 — Solutions & Concentration	Solute/solvent; molarity; dilution; solubility	Calculate molarity/dilution	Quantitative chemistry
Week 7 — Electrochemistry	Oxidation/reduction; oxidation states; batteries; cells	Trace electron transfer	PS1/PS3 applications
Week 8 — Chemistry in Technology & Society	Polymers; batteries; water treatment; green chemistry	Evaluate chemistry applications and tradeoffs	Engineering applications
Week 9 — Module Assessment	Gases; energy; rates; equilibrium; solutions; electrochemistry	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

Chemistry

Module 4 — Applied Chemistry, Environment & Capstone

Week	Topics	Students will be able to	Standards
Week 1 — Carbon Chemistry	Carbon bonding; hydrocarbons; functional groups; polymers; biomolecules	Identify basic organic structures	Chemistry/life-science bridge
Week 2 — Nuclear Chemistry	Decay; radiation; half-life; fission; fusion; applications	Interpret decay models; evaluate applications	Physical science extension
Week 3 — Atmospheric Chemistry	Air pollutants; greenhouse gases; ozone; particulates	Interpret atmospheric chemistry	ESS/PS connections
Week 4 — Water Chemistry	Water quality; dissolved substances; pH; contaminants; treatment	Interpret water-quality data	ESS3/PS1
Week 5 — Climate & Energy Chemistry	Combustion; CO ₂ ; energy resources; chemical energy; efficiency	Analyze emissions and energy tradeoffs	PS3/ESS3
Week 6 — Materials & Engineering	Metals; ceramics; polymers; composites; material properties	Select materials based on properties	Engineering design
Week 7 — Lab & Data Analysis Capstone	Variables; graphs; uncertainty; evidence; CER	Design investigation; analyze data; write CER	SEP emphasis
Week 8 — Chemistry Review & Synthesis	Atomic structure; bonding; reactions; stoichiometry; energy; gases; solutions	Solve unfamiliar chemistry problems	Course synthesis
Week 9 — Final Assessment	Comprehensive chemistry and data assessment	Demonstrate course mastery	Final performance assessment

Module delivery notes. Move from macroscopic observations to particle models and symbolic representations. Introduce quantitative chemistry progressively with strong unit and conservation reasoning.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	HS-PS1, HS-PS3 + engineering connections
Science & Engineering Practices	Questioning; modeling; investigation; mathematics / computational thinking; data analysis; explanation; evidence; design
Crosscutting Concepts	Patterns; cause/effect; systems; energy/matter; structure/function; stability/change
State mapping	California NGSS and Texas high-school science frameworks can be mapped at chapter / question level without changing the core 36-week structure.

Content Production Guidance

For each week, expand the syllabus topic into the established LucidPrep package: ~2-minute opening, objectives, vocabulary, 5 warm-ups with explanations, lecture, 2 ~2-minute videos, AI Instructor presentation, two homework sets of 5, 40-question assessment, answer explanations and mastery/retake handling. Build question banks around phenomena, scenarios, data, models and application.

This 36-week course is designed around **HS-PS2, HS-PS3, HS-PS4 + engineering connections**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	9
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Model motion, forces, energy and momentum mathematically and conceptually.
- Apply Newton's laws and conservation principles.
- Analyze waves, sound, light and information transfer.
- Model electrical and magnetic systems.
- Use equations, graphs, experiments and evidence to explain physical phenomena.

Physics

Module 1 — Motion, Measurement & Forces

Week	Topics	Students will be able to	Standards
Week 1 — Physics & Measurement	SI units; measurement; uncertainty; scalars/vectors; graphs	Use units; distinguish scalar/vector; interpret graphs	SEP/modeling
Week 2 — Describing Motion	Position; distance; displacement; time; speed; velocity	Calculate and interpret motion	HS-PS2
Week 3 — Acceleration	Acceleration; velocity-time and position-time graphs; free fall	Calculate acceleration; interpret graphs	HS-PS2
Week 4 — Forces & Free-Body Diagrams	Net force; gravity; normal; friction; contact/non-contact forces	Draw FBDs; calculate net force	HS-PS2-1/2
Week 5 — Newton's Laws	First/second/third laws; inertia; mass; force	Apply Newton's laws	HS-PS2
Week 6 — Friction & Applications	Static/kinetic friction; drag; engineering applications	Analyze friction and design tradeoffs	HS-PS2
Week 7 — Gravity & Circular Motion	Universal gravitation; mass vs weight; orbits; centripetal acceleration	Model gravitational/circular systems	HS-PS2-4
Week 8 — Motion Investigation	Motion sensors; data tables; graph fitting; evidence	Design and analyze investigation	SEP
Week 9 — Module Assessment	Motion; acceleration; forces; Newton; gravity	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

Physics

Module 2 — Energy, Work & Momentum

Week	Topics	Students will be able to	Standards
Week 1 — Work & Energy	Work; energy; force/displacement; kinetic/potential energy	Calculate work; connect work and energy	HS-PS3
Week 2 — Kinetic & Potential Energy	Kinetic; gravitational; elastic; energy diagrams	Calculate/model energy	HS-PS3-1
Week 3 — Conservation of Energy	Energy transfer; mechanical energy; thermal energy; systems	Track energy and explain conservation	HS-PS3
Week 4 — Power & Efficiency	Power; energy/time; efficiency; machines	Calculate power; evaluate efficiency	HS-PS3
Week 5 — Momentum	Momentum; impulse; force-time	Calculate momentum; explain impulse	HS-PS2-2
Week 6 — Collisions	Elastic/inelastic; momentum conservation; systems	Analyze collisions	HS-PS2-2
Week 7 — Energy & Momentum Engineering	Vehicle safety; crumple zones; airbags; sports; rockets	Apply conservation laws to design	Engineering/PS2/PS 3
Week 8 — Investigation	Experimental design; data; uncertainty; CER	Conduct and explain investigation	SEP/CCC
Week 9 — Module Assessment	Work; energy; power; momentum; collisions	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

Physics

Module 3 — Waves, Sound, Light & Information

Week	Topics	Students will be able to	Standards
Week 1 — Wave Properties	Mechanical/EM waves; amplitude; wavelength; frequency; period; speed	Describe and calculate wave properties	HS-PS4
Week 2 — Sound Waves	Longitudinal waves; pitch; loudness; speed; resonance	Connect frequency to pitch; analyze sound	HS-PS4
Week 3 — Light & EM Waves	EM spectrum; frequency; wavelength; energy; visible light	Compare EM waves and applications	HS-PS4-1
Week 4 — Reflection & Refraction	Reflection; refraction; mirrors; lenses; ray models	Predict optical behavior	HS-PS4
Week 5 — Wave Interactions	Interference; diffraction; superposition; standing waves; resonance	Model wave interactions	HS-PS4
Week 6 — Communication & Digital Information	Signals; analog/digital; encoding; bandwidth; noise	Explain wave-based information transfer	HS-PS4
Week 7 — Waves in Technology	Fiber optics; radio; Wi-Fi; imaging; remote sensing	Connect wave behavior to technology	HS-PS4 applications
Week 8 — Wave Investigation	Measurements; graphs; model fitting; evidence	Measure and model wave behavior	SEP
Week 9 — Module Assessment	Waves; sound; light; optics; information	Demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

Physics

Module 4 — Electricity, Magnetism & Modern Physics

Week	Topics	Students will be able to	Standards
Week 1 — Electric Charge & Fields	Charge; electric force; fields; conductors; insulators	Model electric interactions	HS-PS2
Week 2 — Electric Potential & Circuits	Voltage; current; resistance; series/parallel circuits	Interpret and calculate basic circuits	PS3/PS2
Week 3 — Ohm's Law & Electrical Power	Resistance; Ohm's law; power; household energy	Solve circuit problems; calculate power	Quantitative physical science
Week 4 — Magnetism	Magnetic fields; current; electromagnets; magnetic force	Model magnetic fields and electromagnets	HS-PS2
Week 5 — Electromagnetic Induction	Induction; generators; motors; transformers	Explain energy conversion	PS2/PS3
Week 6 — Modern Physics Foundations	Photons; quantum idea; nuclear energy; relativity overview	Explain limits of classical models	High-school extension
Week 7 — Engineering Design Challenge	Constraints; energy; forces; waves/electricity; optimization	Test and improve a design	NGSS Engineering Design
Week 8 — Physics Review & Synthesis	Mechanics; energy; momentum; waves; electricity; magnetism	Connect principles and solve novel problems	Course synthesis
Week 9 — Final Assessment	Comprehensive quantitative/data/model assessment	Demonstrate course mastery	Final performance assessment

Module delivery notes. Use motion diagrams, free-body diagrams, graphs and system models continuously. Treat equations as models of relationships and anchor them in real systems.

Assessment & Alignment

Assessment Blueprint

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Standards Alignment

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State mapping	California NGSS and Texas high-school science frameworks can be mapped at chapter / question level without changing the core 36-week structure.

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Curriculum Basis & Source Notes

This document is a U.S.-aligned curriculum architecture, not a claim of a single nationwide Grade 9 sequence. NGSS high-school performance expectations are generally written for grades 9–12, while states and districts determine course sequencing.

Primary reference	Next Generation Science Standards (NGSS) high-school Life Science and Physical Science performance expectations, disciplinary core ideas, science and engineering practices, and crosscutting concepts.
State cross-check	California Department of Education — California NGSS, including Grades 9–12 Life Science and Physical Science.
State cross-check	Texas Education Agency — high-school Biology, Chemistry, Integrated Physics and Chemistry, and Physics TEKS.
LucidPrep product alignment	36-week / four-module secondary architecture, weekly instructional components, 40-question assessments, 70% mastery, and review/retake pathway.

Recommended next production step. Add standards metadata to each chapter and question (NGSS PE, DCI, SEP, CCC plus state-framework identifiers) so LucidPrep can support multiple U.S. state alignments without duplicating the course.